

# V3 Multi-Slice Bridge — Mini User Guide

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V3 turns the bridge from a single receiver into a **Flex-style panadapter with multiple "slices"** on one captured window. One SDR, one LO — but several independent narrow receivers living inside the capture, each with its own VFO, mode, and output (WSJT-X over TCI, QMAP, the receiver popout, ...). The capture is wider; the slices roam inside it **digitally** (a per-slice DDC), so a second WSJT-X can chase signals without disturbing the first.

Default behavior is unchanged from V2 — a single slice on the dial. Multi-slice is opt-in via `radio/slice2_enabled=true`.

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## Slices, in order of importance

### Slice 0 — the band-master (TCI 40001)

This is the boss. **Slice 0 is the ONLY slice that moves the SDR's LO / changes bands.** Drive it from your primary WSJT-X (the one on TCI **40001**, CAT 4535, and the bridge's WSJT-X UDP listener). When you change bands here, the whole capture window follows. Everything else is subordinate to Slice 0. It is also the only slice that can **transmit** (TX is a single shared front-end resource).

### Slice 1 — the roamer (TCI 40002)

**Digital-only.** It tunes anywhere inside the captured window by shifting its DDC offset — it **never** moves the LO. Point a second WSJT-X at TCI **40002** and it decodes wherever you put it, with zero effect on Slice 0. It's **RX-only** (no TX — TX belongs to Slice 0) and it remembers its last frequency across restarts.

**Rule of thumb:** *Slice 0 picks the band; Slice 1 (and any further slices) live inside whatever window Slice 0 set.* If Slice 0 jumps to a band that doesn't contain Slice 1's frequency, Slice 1 is flagged out-of-range and muted.

Headline use case: **simultaneous FT8 + FT4** — Slice 0 on 14.074, Slice 1 on 14.080. Both fit one window, one LO, no second radio.

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## Band-snap: what actually moves

The key V3 idea: **band-snap moves the LO park (the capture-window center), not your operating dial.** Your WSJT-X dial stays authoritative — the slice's digital DDC follows it. So:

- Tune **within** the current window → instant, pure-digital, **zero disruption** to the other slices.
- Tune to a **new band** (outside the window) → the LO **re-parks** (a brief retune) to re-center the window optimally for the new band.

That hysteresis means routine QSY never disturbs the roamer — only a real band change does.

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## Park modes — which to pick ( `radio/park_mode` )

Where the LO parks the window center on a band change:

| Mode                   | What it does                                                                                                                           | Use when                                                  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Off</b>             | Park on the dial (legacy V2). Window centered on your operating freq.                                                                  | Simple single-slice, narrow capture.                      |
| <b>EME</b>             | Snap to the EME sub-band centers (50.200 / 144.125 / 432.125 / 902.125 / 1296.125 ...).                                                | EME — so QMAP sees the whole sub-band regardless of dial. |
| <b>Band center</b>     | Per-band, DC-aware: parks the LO a small gap <b>below</b> the band's activity so the digital sub-band sits cleanly above the DC spike. | General HF default.                                       |
| <b>Band lock</b>       | Locks to the band and sizes the <b>capture width</b> to cover it.                                                                      | Wide-capable SDRs you want to see a whole band on.        |
| <b>Day/night block</b> | Parks at the wide-block centers (night ~11.5 MHz covering 160–15 m, day ~19 MHz covering 30–10 m).                                     | Multi-band capture in one window (RX-888 / Pluto-class).  |

## LO placement & the DC spike

Where to put the LO depends on your SDR's **receive architecture**, not just the dial band.

**ZIF / quadrature SDRs** (SDRplay, Pluto, HackRF; and the RTL in normal-tuner / VHF mode) — the **DC spike sits AT the LO** (window center):

- **Narrow HF band** (e.g. 20 m): park the LO just **below the band edge** so the whole band sits one-sided *above* the spike — e.g. LO ~13.97 so DC < 14.000 and the 14.000–14.226 window covers FT8/FT4 above it. *Band-center* mode does this for you.
- **Wide allocation** (e.g. 2 m, 144–146): **center** the LO (~145.000) and accept the DC notch mid-band — 2 m activity (144.1 / 144.174) is narrow and off-center, so the spike falls in dead space.

**Direct-sampling SDRs** (RX-888; RTL HF Q-branch) — **no in-band DC spike** (DC is at true 0 Hz RF, far below HF), so **center freely**.

**The real discriminator is the SDR's receive mode, not the dial frequency.** A 28 MHz-IF transverter op reads ~28 MHz on the dial, but the RTL is back in normal-tuner (ZIF) mode there → the DC spike returns *and* it's an EME/QMAP use case → treat it like VHF (DC-aware park + 256 kHz feed), even though the dial says HF. The bridge keys this off the transverter/direct-sampling state, not the raw frequency.

**VHF/UHF only:** the DC-aware park and the 256 kHz QMAP/Linrad feed both matter at VHF/UHF (where you run EME and a ZIF tuner). On plain HF (direct sampling, no transverter) there's no in-band spike and no QMAP feed — just go as wide as the SDR/CPU/ADC bits allow for the waterfall + slices.

## Quick setup (two slices)

1. Set `radio/slice2_enabled=true` in the bridge INI. **Launch the bridge BEFORE WSJT-X** (TCI sessions don't survive a bridge restart — restart WSJT-X after every bridge restart).
2. Slice 0 WSJT-X → **TCI 40001** (your main, band-master, TX-capable).

3. Slice 1 WSJT-X → **TCI 40002** (the roamer, RX-only).
4. Pick a **park mode** for your operating (EME for EME; Band-center for general HF).
5. **Change bands from Slice 0's WSJT-X** — the window follows, and Slice 1 keeps roaming inside it.

*TX note:* with two TX-capable instances, keying is **first-come, first-served** — the first PTT owns the single TX chain; a second PTT is rejected until release, and an in-progress transmission is never interrupted. In practice the roamer is RX-only, so this rarely comes up.